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A Negative Result on Long-Horizon
Conversational Memory,
and What the Failure Actually Is

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Measurement pass: 2026-09-07 (LoCoMo, zen-embedding-0.6b)

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Abstract

Retrieval over a long conversation fails in a specific and reproducible way, and the standard remedy does not work. On LoCoMo [1], a dense retriever answering *single-hop* questions finds every required turn 78.2% of the time at $k=20$ [measured]. On *multi-hop* questions the same retriever over the same index finds every required turn 22.7% of the time — but finds *at least one* of them 80.1% of the time, which is essentially the single-hop rate. The retriever is not failing to understand the question. It reliably finds a relevant turn and reliably misses the rest.

The textbook fix is to search more than once and let the first result inform the second [3, 4]. We implemented four retrieval policies over one index — a single search, pseudo-relevance feedback, a union over neighbours of the top hits, and a two-deep chain of those — so that the only variable is the policy. **None of them fixes it.** The best gained 0.7 percentage points on multi-hop at double the latency; two were *worse* than a single search; and the one that gained on multi-hop lost 3.1 points on single-hop, so it is not a free choice either.

A diagnostic says why, and it is not subtle. Multi-hop questions need 3.1 gold turns on average (up to 19), and **95.4% of that evidence spans more than one conversation session**. The median gold turn sits at rank 24; the median *worst* gold turn sits at rank 67, beyond $k=20$ for 77.2% of questions and beyond $k=100$ for 43.1%. Additional hops re-search the same similarity space, and a turn that is far away in that space stays far no matter how many times it is queried. The missing signal is not similarity and cannot be recovered by spending more of it. It is *linkage* — which turns concern the same entity across sessions — and that is a structure to be built, not a search to be repeated. We publish this because the negative result is the useful part: it rules out the cheap fix and names the expensive one.

1 Introduction

An agent that holds a relationship with a user over months accumulates a transcript far larger than any context window, so every such system retrieves. The dominant design is a dense retriever over conversation turns: embed each turn, embed the question, return the nearest k . It works well enough that most platforms ship it and report a single recall number.

A single recall number hides the failure. Questions that require *one* fact and questions that require *several* fail in different ways, and averaging them produces a figure that describes neither. We separate them, and the separation is the paper.

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On LoCoMo [1] — ten multi-session conversations of 369–689 turns with human-annotated evidence — a dense retriever at $k=20$ retrieves every required turn for 78.2% of single-hop questions and 22.7% of multi-hop ones [measured]. That gap looks like a retrieval-quality problem and invites a retrieval-quality answer: a better embedding, a reranker, a larger k .

It is not a retrieval-quality problem. The same run reports a second metric that settles it. Scoring *any* rather than *all* — did the retriever find at least one of the required turns — gives 80.7% on single-hop and 80.1% on multi-hop, which are the same number. The retriever understands multi-hop questions exactly as well as it understands single-hop ones. It finds a relevant turn essentially always, and the rest of them essentially never.

That is a precise diagnosis and it points at a known family of fixes. If one search finds part of the answer, search again from what it found: pseudo-relevance feedback [3] in vector space, or the retrieve–reason–retrieve loop of IRCOT [4]. This paper implements four such policies over a single fixed index, measures them, and reports that **they do not work** — and then measures the corpus itself to explain why they cannot.

Contributions

1. A decomposition of dense-retrieval recall on long conversations into *all*-evidence and *any*-evidence, showing the two categories differ only in the former (§3).
2. Four retrieval policies over one index, holding embedding, corpus and scoring fixed so that the policy is the only variable, with the negative result in full including the cases that got worse (§4).
3. A structural diagnostic of the evidence that explains the negative result and predicts it would survive a larger k , a better embedding, or a reranker (§5).
4. A statement of what the measurements do and do not license anyone to build next (§6–§7).

2 Setup

2.1 Corpus and categories

LoCoMo [1] is ten conversations, each spanning multiple sessions over a simulated months-long relationship, 369–689 turns apiece, with questions annotated by category and by the turns that constitute their evidence. We use two categories:

Category 4, single-hop

841 questions answerable from one turn.

Category 1, multi-hop

282 questions requiring evidence from several turns, frequently in different sessions.

Only questions carrying annotated evidence are scored. Retrieval is within a conversation — the correct scope, since a question about one relationship is never answered by another’s transcript.

2.2 Retriever

Deliberately plain, because the paper is about the retrieval *policy* and a sophisticated retriever would confound it: turns embedded with `zenlm/zen-embedding-0.6b`, cosine similarity, brute

force over every turn in the conversation, no approximate index, no reranking, no query rewriting. Brute force costs 1.27 ms per search over a whole conversation [measured], which is small enough that an index would buy nothing at this scale and would add a recall variable we do not want.

Vectors are computed once and cached, so every policy in §4 scores against a byte-identical index. This matters: it means a difference between policies cannot be an embedding difference.

2.3 Two metrics, and why both are needed

For a question with gold evidence set E and retrieved set R_k of size k :

$$\begin{aligned}\text{recall@}k \text{ all} &= \Pr[E \subseteq R_k] \\ \text{recall@}k \text{ any} &= \Pr[E \cap R_k \neq \emptyset]\end{aligned}$$

all is the one that matters for answering: a question needing three facts is not answered by two of them, and a model handed two of three will typically produce a confident wrong answer rather than an admission. **any** is the diagnostic: it separates “the retriever does not understand the question” from “the retriever understands the question and cannot reach all of the answer”. Reporting only one of the two is how the failure in this paper stays hidden.

3 The baseline, and the shape of the failure

	single-hop (841 Q)	multi-hop (282 Q)
recall@5 all	59.1%	10.3%
recall@10 all	69.3%	14.9%
recall@20 all	78.2%	22.7%
recall@5 any	62.2%	54.3%
recall@10 any	71.7%	68.4%
recall@20 any	80.7%	80.1%

Table 1: Dense retrieval on LoCoMo, zen-embedding-0.6b, brute-force cosine [measured]. The bottom row is the one that reframes the problem.

Read the table by column and it says the retriever is bad at multi-hop questions: 78.2% against 22.7% at $k=20$ is a collapse of more than three to one.

Read it by row and it says something different and more useful. On the **any** rows the two columns converge: 80.7% and 80.1% at $k=20$ are within a point of each other. Whatever the retriever is doing, it does it equally well on both categories — it puts a relevant turn in the top 20 about four times in five, regardless of how many turns the question actually needs.

So the failure is not comprehension and not embedding quality. Those would depress **any** as well, and **any** is not depressed. The failure is specifically in *completeness*: the gap between 80.1% and 22.7% — some 57 percentage points — is questions where the retriever found part of the evidence and missed the remainder.

Why more k is not obviously the answer. The single-hop column improves steadily with k ($59.1 \rightarrow 69.3 \rightarrow 78.2$), which is what a retriever that is merely under-budgeted looks like. The

multi-hop **all** column improves too (10.3 $\rightarrow$ 14.9 $\rightarrow$ 22.7) but from so low a base that extrapolating it to a useful number requires a k that defeats the purpose of retrieving. §5 measures how large that k would have to be, and the answer rules the approach out.

4 Four policies, one index

If one search finds part of the evidence, the standard move is to search again from what the first search found. We implemented four policies over the cached index. Every policy sees identical vectors and identical scoring; the retrieval policy is the only variable.

single One search on the question. The baseline.

prf Pseudo-relevance feedback [3]: blend the question vector toward the centroid of its own top 3 hits ($w=0.4$) and search again, keeping the first hop’s best and filling from the second.

multi Search again from each of the top 3 hits; union the results, scoring each neighbour by the product of the seed’s confidence and its own.

chain **multi**, then one further hop from what that found — the retrieve–reason–retrieve shape [4] with similarity standing in for the reasoning step.

Policy	single-hop		multi-hop		ms
	@20 all	Δ	@20 all	Δ	
single	78.2%	—	22.7%	—	1.25
prf	75.1%	−3.1	23.4%	+0.7	2.49
multi	77.9%	−0.3	22.0%	−0.7	5.88
chain	78.1%	−0.1	22.3%	−0.4	8.69

Table 2: Retrieval policy against recall@20 **all**, Δ in percentage points against a single search [measured]. Latency is per search, averaged at $k=20$.

The result is negative, and it is negative in three separate ways.

1. **The best case is noise.** **prf** gains 0.7 percentage points on multi-hop. On 282 questions that is two questions. It is not a result, and reporting it as one would be indefensible.
2. **Two policies made it worse.** **multi** and **chain** — the two that most directly implement “follow the first hit” — both lost ground on multi-hop, at $5\times$ and $7\times$ the latency. Searching from a neighbour pulls in that neighbour’s neighbourhood, which is denser in near-duplicates of what was already found than in the missing evidence.
3. **The one gain is paid for elsewhere.** **prf** costs 3.1 points on single-hop, where the baseline was working. Blending the query toward its own results is a bet that the first hits are on topic; when a single turn is the whole answer, that bet dilutes a query that was already correct.

There is no row in this table anyone should ship. The honest summary is that policy is the wrong knob, and §5 shows it was never going to be the right one.

5 Why no number of hops helps

Rather than try a fifth policy we measured the evidence itself.

Structure of multi-hop evidence (282 questions)	
Gold turns per question, mean	3.1 [measured]
Gold turns per question, max	19 [measured]
Evidence spanning more than one session	95.4% [measured]
Median rank of a gold turn	24 [measured]
Median rank of the <i>worst</i> gold turn	67 [measured]
Worst gold turn beyond $k=20$	77.2% [measured]
Worst gold turn beyond $k=100$	43.1% [measured]

Table 3: The evidence a multi-hop question needs, and where a single similarity search puts it.

Three facts in that table between them close the case.

The evidence is not one neighbourhood; it is several. 95.4% of multi-hop evidence spans more than one conversation session. These are turns separated by simulated weeks, about different immediate topics, phrased differently, related only because they concern the same entity. Nothing about that relation is expressed in the wording, so nothing about it is expressed in a similarity metric over wordings. A second search in the same space is a second look at the same neighbourhood.

The hard turn is not near the easy one. The median gold turn is at rank 24, comfortably retrievable. The median *worst* gold turn is at rank 67. That distance is the entire failure: a question is answered only when the last of its evidence arrives, so $\text{recall}@k$ **all** is governed by the worst-ranked gold turn and by nothing else. Improving the rank of turns that already rank 24 — which is what an extra hop from a top hit does — moves a number the metric does not depend on.

No affordable k rescues it. The worst gold turn is beyond $k=20$ for 77.2% of questions and still beyond $k=100$ for 43.1%. Retrieving 100 turns to answer one question is most of a short conversation; it defeats the purpose of retrieving and it degrades the answer by burying the evidence in context. So the brute-force escape is closed too, and it is closed by measurement rather than by taste.

Putting it together. The retriever finds *a* relevant turn nearly always (§3), because for any multi-hop question at least one piece of evidence does resemble the question. It misses the others because they resemble the question far less — they are linked to it through an entity and a timeline, not through vocabulary. Every policy in §4 is a way of spending more similarity, and similarity is the axis along which the missing evidence is *far*. This is why the negative result is not a tuning failure: no setting of a similarity-based policy has access to the signal that would fix it.

6 What this licenses, and what it does not

Ruled out, on this corpus. Retrieval policy over a dense index — feedback, neighbour expansion, chained hops — as a fix for multi-hop conversational recall. It gained at most 0.7 points, cost

up to $7\times$ the latency, and lost ground where the baseline was working. Also ruled out, by §5 rather than by trial: enlarging k , since 43.1% of questions have evidence beyond rank 100.

Not ruled out, and not tested here. A better embedding would raise both **any** and **all** and we have not measured its ceiling; the diagnostic predicts the gain would be modest, because cross-session evidence is far in *any* similarity space over wordings, but that is a prediction and not a measurement. A cross-encoder reranker over a larger candidate pool addresses the same axis and inherits the same doubt.

What the diagnostic actually points at. The signal the retriever lacks is which turns concern the same entity across sessions. That is a relation to be *constructed* — extract entities as turns arrive, link mentions across sessions, and let retrieval traverse those links rather than only the similarity graph. This is the family that entity-and-graph-structured memory systems occupy [5], and it is the one the measurement endorses.

We are naming a direction, not claiming a result. Building it is a larger project than testing four policies over a cached index, and the honest position at the end of this paper is that we have established what does not work and why, and have not yet built what does. That is the whole content of the claim.

One consequence for anyone shipping today. Report **all**, not **any**, and report the two categories separately. A system quoting 80.1% on multi-hop questions is quoting a true number that describes a system answering 22.7% of them. The gap between those figures is not a rounding difference; it is the difference between a question answered and a confident wrong answer built from two-thirds of the evidence.

7 Limits

One corpus. LoCoMo is ten conversations. It is the standard benchmark for this task [1] and its evidence annotations are what make the **all** metric computable at all, but ten conversations is ten conversations, and the structural facts in §5 — particularly the 95.4% cross-session figure — are properties of *this* corpus. A corpus whose multi-hop evidence clustered within sessions would very likely behave differently, and the policies might help there.

One embedding. Every row uses **zen-embedding-0.6b**. The comparison between policies is sound because they share an index, but the absolute recall numbers are that model’s, and a stronger embedding would shift the whole table. What would *not* shift is the relationship the paper is about: **any** exceeding **all** by a wide margin is a statement about evidence distribution, not about embedding quality.

Similarity standing in for reasoning. Our **chain** policy follows IRCOT’s shape [4] with one important substitution: IRCOT interleaves a language model’s reasoning between retrieval steps, and we substitute vector similarity for that step. This makes the policy thousands of times cheaper and it is a fair test of “does searching again from what you found help”, but it is *not* a test of IRCOT. A model that reads the first hop and writes a genuinely new query has access to exactly the entity linkage §5 identifies as missing. We expect it to do better; we did not measure it; and it costs a model call per hop, which is a different economic proposition from 8.69 ms.

Retrieval only. We measure whether the evidence was retrieved, never whether a model answered correctly given it. End-to-end accuracy depends on the reader as much as the retriever, and conflating the two is how retrieval failures get attributed to models. The scope here is deliberate and narrow.

8 Related work

LoCoMo [1] established the long-horizon conversational memory task and supplies the per-question evidence annotations that make an **all**-evidence metric computable; without them only **any** is measurable, and §3 is the argument that **any** alone is misleading.

Dense passage retrieval [2] is the substrate every policy here runs on. Pseudo-relevance feedback long predates it — Rocchio’s method [3] reformulates a query toward its own top results, and our **prf** policy is that idea in vector space. IRCOT [4] interleaves retrieval with a model’s chain of thought, which is the shape our **chain** policy imitates with similarity substituted for the reasoning step; §7 is explicit that this makes our result a bound on cheap hops rather than a refutation of IRCOT.

Entity-and-graph-structured memory [5] is the family §6 points at: it constructs the cross-session linkage that our diagnostic identifies as the missing signal, instead of trying to recover it from similarity. This paper’s contribution to that line is negative and supporting — evidence, on a standard benchmark, that the cheaper alternative does not work and cannot be made to.

9 Conclusion

Dense retrieval over a long conversation does not misunderstand multi-hop questions. It answers them partially: at $k=20$ it finds at least one required turn 80.1% of the time and all of them 22.7% of the time.

Four retrieval policies over one fixed index — feedback, neighbour expansion, and a two-deep chain — move that 22.7% by at most 0.7 percentage points, and two of the four move it backwards. The reason is structural: 95.4% of multi-hop evidence spans sessions, the median worst-ranked gold turn sits at rank 67, and 43.1% of questions have evidence past rank 100. Every policy tested spends more similarity, and similarity is precisely the axis along which the missing turns are distant.

The fix is not a better search over the same space. It is a different relation — entity linkage across sessions — built as the conversation arrives and traversed at retrieval time. We have not built it. We have established, on a standard benchmark with a reproducible script, that the cheap alternative does not work, and we would rather publish that than a tuning result of two questions out of 282.

Reproducing

```
curl -sL https://raw.githubusercontent.com/snap-research/locomo/main/\
data/locomo10.json -o locomo10.json
node brain.mjs      # baseline; caches vectors to brain-vectors.json
node hops.mjs      # the four policies, from the cache
node evidence.mjs  # the structural diagnostic
```

The scripts are in `hanzoai/cloud` under `bench/brain/`.

References

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